



Examining influencing factors in measuring mean length of utterance: A case study of Chinese-speaking older adults with cognitive decline

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ABSTRACT

Linguistic changes in later life serve as observable indicators of age-related cognitive decline, with mean length of utterance (MLU) being a widely adopted metric for assessing syntactic complexity in aging populations. However, the reliability of MLU depends on utterance segmentation consistency, a methodological challenge that remains underexplored in Chinese, where standardized guidelines are lacking and phenomena such as pro-drop and cohesive repetition systematically obscure boundary judgments. This study presents a methodological investigation into how segmentation variability influences MLU values. Using narrative and descriptive speech samples from Chinese older adults with differing cognitive statuses, three MLU variants were derived from the transcripts: MLU_{Raw}, MLU adjusted for pro-drop (MLU_{PPD}), and MLU adjusted for cohesive repetition (MLU_{CHS}). Inter-annotator agreement in utterance boundary judgments was assessed via accuracy, while Pearson Correlation and Bland-Altman analysis quantified cross-method agreement. Results showed that without explicit segmentation guidelines, inter-annotator agreement was quite low, indicating high subjectivity. Bland-Altman analysis showed significant MLU variability between calculation methods. Task complexity and cognitive status critically moderated MLU validity. MLU_{PPD} amplified differences between cognitively impaired and healthy older adults during narrative tasks. In contrast, MLU_{Raw} and MLU_{CHS} demonstrated high agreement, supporting their stability for basic language assessment. The findings demonstrate that MLU is not a fixed property of speech but a construct highly sensitive to preprocessing decisions, task demands, and annotator interpretation. This study underscores the necessity for evidence-based, linguistically motivated segmentation protocols for Chinese and provides a methodological framework for evaluating measurement consistency. It argues that systematic methodological investigation including explicit reporting of segmentation criteria and multi-method agreement analyses is a prerequisite for advancing MLU as a valid measure in lifespan linguistics and clinical applications.

1. Introduction

Changes in linguistic capacity serve as a salient and observable indicator of age-related cognitive decline in the daily functioning of older adults, which constitutes a vital resource for promoting individual or population health (Huang, 2021). Rigorous linguistic research can actively advance scientific exploration and improve human care, thus contributing to the health of the whole society. Investigations into this domain hold significant potential to augment the theoretical framework of linguistics in the lifespan perspective. Specifically, such research promotes the knowledge of lifespan linguistics by facilitating the comprehensive description, analysis, and interpretation of linguistic phenomena in later stages of individuals' lives. Such research outcomes,

therefore, constitute a critical perspective for investigating human language development. Significant advancements in the study of linguistic change across the lifespan has thus converged on a multidisciplinary research paradigm, necessitating the integration of diverse theoretical frameworks and methodological approaches (Huang, 2024).

Recently, some comprehensive and in-depth investigations have been conducted into the impact of cognitive decline on linguistic features across multiple levels, encompassing lexical, syntactic, semantic, and pragmatic domains and their practical use in the clinical screening for cognitive function (Dai, 2022; Huang et al., 2022; Huang & Che, 2023; Huang et al., 2024; Yih et al., 2025). In recent years, particular emphasis has been placed on examining the effects of cognitive impairment, including Alzheimer's disease (AD), on syntactic and

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semantic dimensions in older adults. This focus stems from prior research suggesting that syntactic deterioration typically manifests only during the moderate-to-severe stages of AD (Szatloczki et al., 2015). Consequently, whether similar patterns manifest within the Chinese elderly population presents a significant question meriting investigation.

1.1. The development of MLU

In the interdisciplinary studies of language and health, mean length of utterance (MLU) is often selected as the key feature for exploring syntactic phenomena. MLU is a commonly used simple measure of grammatical development because almost every new kind of knowledge increases length, so the average length of utterance can serve as a useful indicator to measure syntactic development (Brown, 1973). Brown first introduced the use of morphemes as the unit for MLU and demonstrated that the average number of morphemes per utterance steadily increases with age based on longitudinal data from three English-speaking children. Many studies have used MLU as an indicator of syntactic development and language disorders across populations (de Villiers & de Villiers, 1973; Miller & Chapman, 1981; Tager-Flusberg et al., 1990; Zackheim & Conture, 2003). Although its validity beyond early childhood has been questioned (Klee & Fitzgerald, 1985), and its utility varies across populations, a growing body of research has examined MLU in the context of aging and cognitive decline. In aging populations, MLU serves as a measure of syntactic complexity in connected speech and has been consistently linked to cognitive status. Specifically, MLU has shown promise as a marker of cognitive decline in aging based on several lines of evidence.

First, longitudinal studies have demonstrated that grammatical complexity and mean utterance length decline with age, and these declines are markedly accelerated in Alzheimer's disease (Kemper et al., 2001b). Using longitudinal connected speech data from autopsy-proven Alzheimer's patients, Ahmed et al. (2013) showed that reduced MLU tracks neuropathological progression, validating it as a reliable indicator of disease advancement over time. Second, MLU decline can be detected even in the very early stages of cognitive impairment. Mueller et al. (2018) found that individuals with very mild cognitive impairment already produce shorter and syntactically simpler utterances during narrative tasks, suggesting that MLU reduction precedes clinical diagnosis. Third, MLU has been successfully employed in machine learning classifiers to distinguish Alzheimer's patients from healthy controls with considerable accuracy (Fraser et al., 2015). Fourth, from a lifespan perspective, lower grammatical complexity in early adulthood has been shown to predict the risk of Alzheimer's disease in late life (Kemper et al., 2001a), indicating that syntactic skills indexed by MLU may reflect cognitive reserve. Furthermore, MLU has been found to correlate with cognitive flexibility under high load (Fleming, 2007) and with syntactic comprehension in clinical populations (Sazegar Nejad et al., 2025), and is modulated by cognitive reserve factors such as education and social engagement (Dai, 2022).

Collectively, these findings position MLU as a potentially valuable behavioral marker for cognitive-linguistic health in aging. Nevertheless, the utility of MLU in both research and clinical settings hinges critically on measurement consistency. If the same speech sample yields substantially different MLU values depending on how utterance boundaries are determined, the comparability of findings across studies and the reliability of individual-level classifications will be seriously compromised. This concern leads directly to the methodological challenge of utterance segmentation.

1.2. The controversy of utterance segmentation

In linguistic research, utterance segmentation is a critical step for calculating measures such as MLU, while there are no universally accepted standardized utterance segmentation rules for MLU calculation

at present. Miller and Chapman (1981) took "terminal intonation contour" as the main basis for segmenting utterances, determining the end of an utterance by identifying the termination patterns of intonation (such as prosodic features like falling tone, rising tone, or flat tone). However, this method has a 10–15% inter-researcher disagreement, indicating that relying solely on intonation cues is insufficient to fully resolve the problem of defining utterance boundaries. On this basis, Leadholm and Miller (1992) added the provision that a pause of more than 2 seconds should be regarded as an utterance boundary. In addition, for long sentences containing multiple coordinate conjunctions (such as *and*), the study clearly required them to be segmented into multiple independent utterances, with each utterance containing at most one clause, in order to avoid analysis bias caused by excessive verbosity. Lund and Duchan (1993) and Owens (1999) suggested segmenting utterances in combination with grammatical structures, regarding word groups or phrases that conform to the definition of sentences as complete utterances. Owens (1999) further proposed using inhalation as an auxiliary cue: when a child shows an obvious inhalation behavior, it can be used as the boundary between preceding and following utterances. These supplementary rules aim to improve the accuracy of utterance segmentation through multi-dimensional criteria, but there are still controversies in practical operations (such as the treatment of vocatives and yes/no responses), suggesting that a more standardized segmentation process needs to be established to ensure the reliability of MLU calculation.

For MLU calculation involving Chinese, its segmentation criteria must integrate semantic, pragmatic, and syntactic information, making it difficult to apply English segmentation standards directly. As an analytic language, Chinese lacks explicit morphological markers, making it more challenging to establish uniform segmentation standards. Unlike Indo-European languages such as English, where segmentation can rely on syntactic markers, Chinese segmentation often depends more on semantic and pragmatic information (Zhang et al., 2022).

The practical implementation of utterance segmentation faces several challenges. Real language data is highly complex, involving phenomena such as ellipsis, repetition, self-repair, and parenthetical insertions, necessitating detailed segmentation rules. Yet, many empirical studies fail to explicitly articulate their segmentation principles, compromising the comparability and reproducibility of their findings. This lack of clarity in segmentation principles makes it difficult to directly compare results across studies. Segmentation criteria are inherently subjective, and different approaches can lead to varying MLU calculations, influencing research conclusions. There is a lack of cross-linguistic and cross-cultural research, with most studies centered on English or other high-resource languages, leaving low-resource languages and cross-cultural contexts underexplored. Given the significant variations in utterance boundary markers across languages and cultures, further cross-linguistic studies are needed to validate the universality of existing methods (Evans & Levinson, 2009).

Two linguistic phenomena are of particular relevance to utterance segmentation in Chinese and may systematically affect MLU values: pro-drop and cohesive repetition. Pro-drop refers to the linguistic phenomenon where a language allows pronouns (particularly subjects or objects) to be syntactically omitted or "dropped" from a sentence when their reference can be recovered from context (Huang, 1989). The term specifically denotes the occurrence of a null pronominal element (pro) in positions where overt pronouns would appear in non-pro-drop languages like English. Chomsky (1965) distinguishes two null elements: PRO, a pronominal anaphor restricted to ungoverned subject positions of non-finite clauses, and pro, a pure pronominal that may appear in governed positions (e.g., subject/object of finite clauses) in pro-drop languages like Italian or Spanish. The Identification Hypothesis governs the distribution of pro: a pronoun may drop only if its reference (e.g., person, number, gender) is recoverable from grammatical features elsewhere in the sentence. Huang (1989) argues that Chinese, despite lacking verbal agreement, exhibits pervasive pro-drop (allowing null

subjects and objects in finite clauses). This pro-drop phenomenon often appears in clauses that are phonetically separated by pauses but grammatically ambiguous due to null pronouns. For example, they either share the same subject with the preceding clause or involve switch-reference, where the logical subject of the subsequent clause shifts to the object or another element from the prior clause. Such cases may cause structural ambiguity and complicate sentence segmentation which could lead to different utterance counts. Cohesive repetition (CHS) describes the phenomenon where content from a prior utterance is repeated (often with verbal markers) before introducing new information, which artificially inflates utterance length and distorts MLU calculations. This phenomenon is common in spontaneous Mandarin Chinese speech, as illustrated by the following examples:

- (1) 这个人晾了衣服，晾了衣服在滴水。
Zhège rén liàng le yīfu, liàng le yīfu zài dīshuǐ.
'This person hung out the laundry, hung out the laundry and it is dripping.'
- (2) 他们到外面去郊游了，郊游了以后小孩子也带去了。
Tāmen dào wàimiàn qù jiāoyóu le, jiāoyóu le yǐhòu xiǎoháizi yě dài qù le.
'They went on an outing, went on an outing and then took the children along.'

These examples show how CHS inflates utterance length without adding core propositional content.

The importance of segmentation standardization is further underscored when examining existing empirical studies on aging and language. Although studies have reported associations between MLU and age-related changes in speech breathing (Huber, 2008), acoustic adaptation (Hazan et al., 2018), and cognitive reserve (Dai, 2022), few have explicitly articulated their utterance segmentation criteria.

When such criteria are reported, they vary considerably across studies: some rely primarily on prosodic cues, others incorporate pause duration thresholds or syntactic heuristics, and many remain silent on how they handle phenomena such as repetitions, self-repairs, and pro-drop constructions. Tomas and Dorofeeva (2019) provide detailed manual guidelines: utterance boundaries are determined primarily by prosodic and semantic completeness, with a 500-ms pause as an auxiliary cue, and long but coherent interruptions are treated as a single utterance. Shriberg et al. (2000) propose an automatic method using prosodic features (pause duration, pitch reset, pre-boundary lengthening) within a decision tree/HMM framework to detect sentence and topic boundaries. In contrast, many other studies (e.g., Brent & Cartwright, 1996; Cutler et al., 1997; Stolcke et al., 2000; Swingley, 2005; Evans & Levinson, 2009; Allen & Dench, 2015) rely on pre-segmented corpora or focus on word, syllable, or morpheme level analysis without specifying segmentation rules for utterances.

This lack of transparency and consistency makes direct cross-study comparison of MLU values difficult, if not impossible. Consequently, it remains unclear whether discrepancies in findings across studies reflect genuine population differences or simply methodological artifacts arising from different segmentation choices. This uncertainty provides a strong motivation for the present methodological investigation.

1.3. The present study

Given that segmentation variability can alter MLU values and that no consensus exists for Chinese, the present study empirically examines the impact of different clause segmentation schemes on MLU in older Chinese adults, and assesses the consistency of utterance boundary judgments when strict guidelines are absent. The research questions are as follows:

- (1) To what extent do utterance segmentation judgments vary across annotators, and how does this variation affect the calculation of MLU?
- (2) How do different clause segmentation schemes influence the discriminant validity and reliability of MLU in Chinese older adults?

Based on the literature reviewed above, the following hypotheses are proposed. Regarding the first research question, it is hypothesized that in the absence of strict segmentation guidelines, inter-annotator agreement on utterance boundary judgments will be only moderate or low, reflecting the inherent ambiguity of prosodic and semantic cues in Chinese connected speech. Regarding the second research question, it is hypothesized that different clause segmentation schemes will yield significantly different MLU values. Furthermore, the importance of these differences is predicted to be large enough to affect the classification of individuals' syntactic complexity levels, thus underscoring the importance of segmentation standardization in Chinese language research for older adults.

2. Methodology

2.1. Participants

This study employed a comparative design involving a cognitively impaired group (the MCI group, $n = 26$, 5 males and 21 females) and a healthy control group (the healthy group, $n = 24$, 7 males and 17 females), totaling 50 participants aged over 60 recruited from a community in Yangpu District, Shanghai. The majority of participants presented with common chronic conditions in older adults, including hypertension, diabetes, and hyperlipidemia. Participants in the MCI group were clinically diagnosed with cognitive impairment through the Montreal Cognitive Assessment-Basic (MoCA-B, Chinese version) test administered by doctors or qualified examiners from professional medical institutions. According to the *Dementia and Cognitive Disorders Group of the Chinese Society of Neurology* (2025), the education-adjusted MoCA-B cutoff scores for MCI are as follows: (1) a score of 19 or lower for individuals with 6 or fewer years of education; (2) 22 or lower for those with 7–12 years of education; (3) 24 or lower for those with more than 12 years of education. Only participants confirmed by doctors or qualified examiners from professional medical institutions as meeting all MCI criteria were assigned to the MCI group. The healthy group comprised age-matched individuals with normal cognitive functioning.

To ensure that this study is methodologically rigorous, individuals with hearing or visual impairments, as well as those with aphasia, were excluded from both groups. According to Table 1, Mann-Whitney U nonparametric tests were used to confirm that there was no significant intergroup difference in age ($p > .05$), while there was a statistically significant difference in cognitive performance between the two groups ($p < .001$). The MCI group demonstrated significantly lower cognitive

Table 1
The demographic characteristics of the participants.

Variable	MCI Group ($n = 26$)	Healthy Group ($n = 24$)	Test Statistic	p value
Gender (Female)	21	17	$\chi^2 = 0.675$	$> .05$
Age (Years)	71.0 (66.8, 76.8)	70.5 (68.0, 74.8)	$U = 287.5$	$> .05$
Education (Years)	9.0 (9.0, 12.0)	11.5 (9.0, 12.0)	$U = 237.5$	$> .05$
MoCA-B score	17.5 (14.0–20.0)	25.5 (24.0–27.0)	$U = 6.5$	$< .001^{***}$

Note: Gender was presented as female frequency and compared by χ^2 test. Continuous variables (age, years of education, MoCA-B score) were shown as median (IQR) and compared by Mann-Whitney U test.

scores, thereby establishing validated group differences for subsequent experimental analyses.

2.2. Speech tasks and procedure

This study employed two sets of visual stimuli with moderate information complexity for the speech tasks. The first task was the *Picnic Scene* from the *Western Aphasia Battery*. The scene contains multiple characters, objects and actions, requiring the speaker to organize diverse verbal descriptions. The second task used a six-picture sequence from the *Wanderings of Sanmao*. This culturally familiar scene was selected to prompt naturalistic narrative language from older Chinese speakers.

Visual stimuli are recognized for their efficacy in language disorder assessment, and Goodglass and Kaplan (1983) established that picture description tasks provide standardized visual input, elicit contextualized language production, and minimize contextual interference, making them ideal for studying linguistic characteristics in cognitively impaired older adults.

The speech tasks were conducted individually in quiet environment without distractions. Both pictures were presented individually on printed paper. After explaining the procedure of this study, the examiner presented the images one by one using standardized instructions “Please describe the story in this picture using complete sentences” for the *Picnic Scene*, and “These six pictures tell one whole story. Beginning with this image, please tell the entire story to me” for the *Wanderings of Sanmao*. If the participant did not respond within 10 s, the experimenter would point to the picture to prompt the participant. Also, if the participant described fewer than two sentences, they would be prompted again, and would be asked about activities, time, and objects in the picture. If there was no new information after further prompting, or the participant clearly indicated that he/she was finished, then the task would be terminated.

Participants’ verbal responses were audio-recorded. After each task, audio files were labeled with the participants’ Chinese initials and cataloged for analysis. A total of 47 *Picnic Scene* recordings and 46 *Wanderings of Sanmao* recordings were collected; a small number of files were excluded due to technical damage.

All audible speech was transcribed manually by trained research assistants, following a verbatim convention that preserved every vocalization, including filled pauses (e.g., *en* 嗯, *e* 呃), repetitions, false starts, self-corrections, and incomplete words. No material was removed from the transcripts prior to the initial word segmentation. Punctuation was omitted to avoid imposing sentence boundaries prematurely. Each transcript thus constituted a faithful orthographic record of the participant’s full spoken output.

For the calculation of Raw MLU, the complete, uncleaned transcripts were submitted to the Python package Jieba¹ for word segmentation. Jieba was applied to the entire transcript without any prior deletion of disfluencies or filled pauses. The resulting word sequences served as the input for all subsequent MLU calculations.

Subsequently, two manual adjustments were applied to address specific linguistic phenomena that could distort MLU: pro-drop (PDP) and cohesive repetition (CHS). For PDP adjustment, clauses that were phonetically separated by pauses but grammatically ambiguous in terms of sentence boundaries were manually reviewed. For instance, null-subject clauses sharing a referent with the preceding clause were manually reviewed and, when appropriate, merged into a single utterance. For CHS, instances where content from a prior utterance was repeated with a cohesive marker before new information were edited to remove the repeated material. After each manual operation, the transcripts were re-segmented with Jieba and MLU was recalculated, yielding MLU_{PDP} and MLU_{CHS} respectively. All manual annotations were performed by two independent annotators, with disagreements resolved

through discussion.

This three-variant approach (Raw, PDP, CHS) allowed us to examine how different segmentation and editing decisions affect MLU values. All figures were generated using R.²

2.3. Measures

As discussed in Section 1.2, pro-drop and cohesive repetition are two linguistic phenomena that may systematically bias MLU calculation in Chinese connected speech. To empirically evaluate their impact, we operationalized three MLU variants derived from the same transcripts through different preprocessing procedures.

Raw MLU Calculation: The MLU_{Raw} was computed directly from the full, uncleaned word-segmented transcripts. This variant preserved all original disfluencies, filled pauses, repetitions, and pro-drop structures, thus reflecting the participant’s unedited spoken output.

Pro-drop (PDP) MLU Calculation: To address the potential fragmenting effect of pro-drop, two independent annotators manually reviewed the transcripts and merged clauses that were phonetically separated but grammatically dependent on a shared null subject. After merging, the edited transcripts were re-segmented with Jieba and MLU was recalculated, yielding MLU_{PDP}.

Cohesive Repetition (CHS) MLU Calculation: To address the potential inflating effect of cohesive repetition, the same two annotators identified and removed instances where content from a prior utterance was repeated with a discourse marker before new information was introduced. The edited transcripts were then re-segmented and MLU was recalculated, yielding MLU_{CHS}.

The three MLU variants (Raw, PDP, CHS) were computed for each participant for each task, producing two complete datasets for subsequent statistical analysis.

2.4. Statistical analysis

To address the first research question, we adopted a segmentation-based evaluation method commonly used in natural language processing tasks such as Chinese word segmentation and named entity recognition (Zhang & Teng, 2021). In this framework, utterance segmentation is treated as a sequence labeling task. The transcribed texts were first divided into segments based on audible pauses identified during transcription, which were marked with temporary separators. The boundaries between these segments served as potential utterance boundaries. Next, two master’s students in linguistics independently annotated each boundary. Given the lack of consensus on the definition and delimitation of “utterance”, the two annotators conducted the annotation based on their understanding of the notion. Each potential boundary was assigned one of two labels, ‘True’ or ‘False’. If the annotator deemed that two adjacent clauses should be merged into a single utterance, the boundary was labeled as ‘False’; otherwise, it was labeled as ‘True’. Finally, a given text can be represented as a sequence of potential labels, as shown in Fig. 1.

To assess inter-annotator agreement, the two resulting sequences of boundary labels were then compared. We considered three candidate evaluation measures (F-score or F1, Cohen’s kappa, and accuracy), and ultimately selected accuracy. For F-score, although it is the most commonly used measure in evaluating sequence labeling tasks (Tjong Kim Sang & De Meulder, 2003), it was not adopted here due to its asymmetric nature that typically requires the specification of a gold standard. This is feasible in NLP tasks with a predefined correct answer, but not applicable when evaluating inter-annotator agreement, where no annotation can be assumed to be the ground truth. Cohen’s kappa, which is widely used in corpus linguistics for measuring inter-annotator reliability (Levshina, 2015), is also inappropriate in this context. This is

¹ <https://pypi.org/project/jieba/>

² <https://ggplot2.tidyverse.org/>

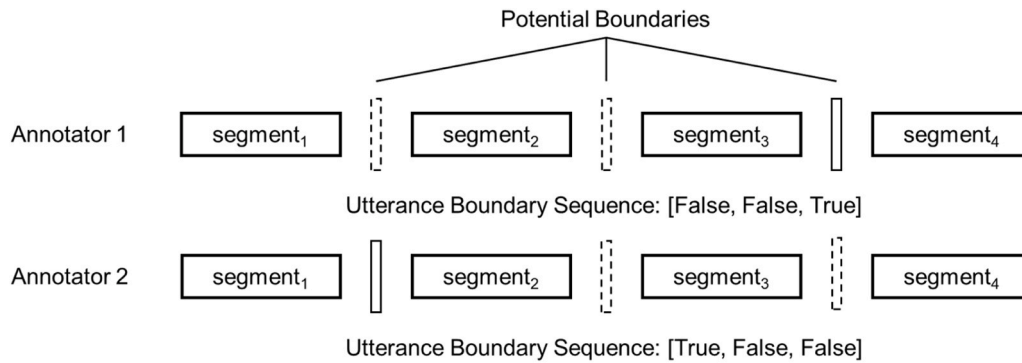


Fig. 1. A diagram illustrating the agreement between the two annotators in utterance segmentation.

because kappa becomes zero whenever one annotator assigns the same label throughout the sequence, a situation that frequently occurs in utterance segmentation, where an annotator may mark all potential boundaries as true boundaries. In such cases, kappa fails to provide meaningful information. As for accuracy, while class imbalance may theoretically result in inflated value, this issue did not arise in our pilot experiments. Therefore, we ultimately adopted accuracy (ACC) as the evaluation measure, as it better reflects the consistency between the two annotators in utterance segmentation. Values approaching 1 indicate higher agreement, while values closer to 0 suggest substantial disagreement between annotators.

To address the second research question, we adopted Pearson correlation coefficients and Bland-Altman difference plot to further analyze following the approach Tomas and Dorofeeva's research in 2019. Significant differences in the Pearson correlation coefficients or in the Bland-Altman difference plot between the two topics were interpreted as evidence that MLU values are sensitive to task-specific factors, which would indicate limited cross-task validity of MLU as a linguistic measure. Pearson correlation coefficients of MLU obtained from the two different datasets were analyzed. The three MLU variants in each dataset were calculated in pairs, aiming to examine the effect of different clause segmentation schemes on MLU. Also, Bland-Altman difference plots of MLU obtained from the two different datasets were analyzed. Bland-Altman analysis is a statistical method used to evaluate the agreement between two different measurement techniques by analyzing the differences between their measurements. This analysis helps determine if two methods can be used interchangeably or if systematic biases exist that might affect research conclusions. Each dataset contained measurements from two methods applied to the same participant. The analysis included calculating mean differences, standard deviations of differences, 95% confidence intervals, and coefficients of repeatability. Statistical significance of mean differences was assessed using *t* tests.

3. Results

3.1. Inter-annotator agreement in utterance segmentation

To explore the utterance segmentation consistency of the text, we invited two master's students majoring in linguistics to segment the collected texts into utterances and then calculate the accuracy (ACC) metric of each text. The results showed that there was a certain difference in the overall accuracy rate of the two groups of tasks: the overall ACC of the narrative task was 42.17%, while that of the descriptive task was slightly higher, which was 44.52%, as demonstrated in Fig. 2.

From the perspective of individual performance, the accuracy rate of some participants in the two tasks showed a similar trend, while others showed obvious differences in task dependence. The extreme values of some subjects in the descriptive task (the highest value of 71.43%) were slightly higher than those of the narrative task (the highest value of 68.18%), but both showed large internal variation.

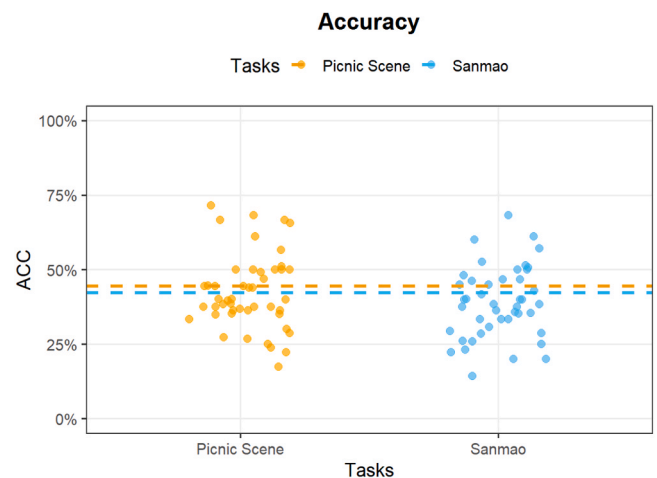


Fig. 2. Accuracy (ACC) rate of the two tasks.

The Picnic Scene, as a concrete scene descriptive task, may activate a more stable language model, while *the Wanderings of Sanmao*, as a narrative task, requires a higher level of semantic integration, resulting in fluctuations in the performance of some participants. Participants with high consistency in ACC rate may adopt adaptive coding strategies, while participants with low consistency may be limited by working memory capacity, making it difficult to maintain stability across tasks. The variability of sentence consistency rate or the sensitivity of task design to language output should be further verified in combination with eye-tracking or neuroimaging data.

3.2. Correlations between MLU calculation methods

In addition, Pearson correlation coefficients analysis was used in the two tasks to provide valuable insights into the relationships between different MLU calculation methods and their implications for language development research. As shown in Table 2, the strong correlation between MLU_{Raw} and MLU_{CHS} (0.968) in the narrative task suggests that these two approaches yield consistent results, which is crucial for maintaining reliability across different analytical techniques. However, the weaker correlations involving MLU_{PDP} (0.438 with MLU_{Raw} and 0.358 with MLU_{CHS}) indicate that this method may capture utterance

Table 2
Pearson correlation coefficients of *the Wanderings of Sanmao*.

	MLU_{Raw}	MLU_{PDP}	MLU_{CHS}
MLU_{Raw}	1.000		
MLU_{PDP}	0.438	1.000	
MLU_{CHS}	0.968	0.358	1.000

properties that differ substantially from those indexed by MLU_{Raw} and MLU_{CHS} . This divergence likely reflects the fact that clause merging based on pro-drop alters the utterance count in a way that emphasizes syntactic dependency and discourse cohesion, rather than simply measuring surface utterance length. Therefore, MLU_{PDP} should be viewed not as an unstable variant, but as a measure that is sensitive to a distinct dimension of linguistic structure.

In contrast, the descriptive task shows remarkably high correlations across all method pairs, ranging from 0.944 to 0.999 in Table 3. Such strong agreement suggests that, for this particular task, the choice of MLU calculation method has minimal impact on the results. This consistency is advantageous for research reproducibility and allows for greater flexibility in methodological approaches. The high correlation between MLU_{Raw} and MLU_{CHS} (0.999) is particularly noteworthy, as it indicates that the adjustment for cohesive repetition does not alter the fundamental measurement of utterance length in this context.

3.3. Bland-Altman analysis of method agreement

The Bland-Altman analysis was also used to figure out the consistency and potential interchangeability of different measures. The three pairs of MLU calculations were analyzed in different tasks and groups.

According to Table 4, in the narrative task ($n = 46$), Bland-Altman analysis revealed systematic differences among the three MLU calculation methods. The significant mean difference of -3.529 ($p < .001$) between MLU_{Raw} and MLU_{PDP} indicates a substantial systematic bias where the first method tends to produce much lower values than the second method. The extremely wide 95% limits of agreement (-10.731 – 3.673) suggest that the measurements from these two methods can vary greatly, making them unsuitable for interchangeable use in this context. Also, it presents a more refined picture with a mean difference of -0.316 and a significant p-value ($< .001$) between MLU_{Raw} and MLU_{CHS} . While the bias is smaller, the 95% limits of agreement (-1.500 – 0.868) still indicate a notable range of differences between the methods. This suggests that although the methods show some level of agreement, there are still systematic differences that could affect research conclusions, especially in precise calculation. For MLU_{PDP} and MLU_{CHS} , it shows a significant mean difference of 3.213 ($p < .001$), with the first method producing higher values than the second.

For the MCI group and healthy group, the MLU_{PDP} measure exhibited fundamental divergence from other calculation approaches, as it demonstrates a significant negative bias compared to MLU_{Raw} ($p < .05$), while showing a positive discrepancy against MLU_{CHS} ($p < .05$). This bidirectional deviation reveals MLU_{PDP} 's unique sensitivity to syntactic fragmentation induced by pronoun drops, particularly within the cognitively impaired group, where these discrepancies expand with 95% limits of agreement (LoA) widening by 96% in Fig. 3.

Conversely, MLU_{Raw} and MLU_{CHS} methods display high concordance: their differences remain minimal ($p < .05$) with narrow 95% LoA. This cross-group stability establishes them as reliable measures for foundational language assessment. Crucially, the variability amplitude of MLU_{PDP} measure related differences strongly negatively correlates with cognitive levels with its fluctuation range directly quantifying the instability of grammatical boundary judgments.

In the descriptive task ($n = 47$), Bland-Altman analysis revealed a distinct methodological consistency pattern compared to the narrative task: MLU_{Raw} and MLU_{CHS} methods demonstrated near-perfect agreement ($p > .05$) with 95% LoA compressed to -0.122 – 0.102 and a

Table 4

The Bland-Altman Analysis of the Wanderings of Sanmao.

	MLU_{Raw} & MLU_{PDP}	MLU_{Raw} & MLU_{CHS}	MLU_{PDP} & MLU_{CHS}
Samples	46		
Mean (1 st method)	5.924	5.924	9.453
Mean (2 nd method)	9.453	6.24	6.24
Mean (Difference)	-3.529	-0.316	3.213
S. D. (Difference)	3.675	0.604	3.815
95% CI	$-4.620 \sim$	$-0.496 \sim$	$2.080 \sim 4.346$
(Mean of Difference)	-2.438	-0.137	
95% CI (Difference)	$-10.731 \sim$	$-1.500 \sim 0.868$	$-4.265 \sim$
	3.673		10.691
<i>t</i>	-6.514	-3.548	5.712
<i>p</i>	$< .001$	$< .001$	$< .001$
Coefficient of Repeatability	9.929	1.325	9.714

repeatability coefficient as low as 0.112, indicating that repetition redundancy had negligible impact on utterance length. Conversely, MLU_{Raw} and MLU_{PDP} showed significant negative bias ($p < .05$), elevating MLU by 8.0%, while MLU_{PDP} and MLU_{CHS} showed positive discrepancy ($p < .05$), revealing inherent tension between pro-drop correction and repetition-redundancy processing. Crucially, all variability was substantially suppressed, as the LoA of MLU_{Raw} and MLU_{PDP} decreased to -1.441 – 0.663 , as well as the LoA of MLU_{PDP} and MLU_{CHS} to -0.692 – 1.449 . The repeatability coefficients were merely 13% of those in the narrative task, proving that simple scene descriptions drastically diminish grammatical phenomena's impact on MLU calculation.

According to Table 5, the significant mean difference of -0.389 indicates a systematic bias between MLU_{Raw} and MLU_{PDP} . The wide 95% LoA (-1.441 – 0.663) suggests substantial variability in the differences between measurements, making them unsuitable for interchangeable use in this context. It also presents a sharp contrast with a negligible mean difference (-0.01) and a non-significant p-value (0.226), indicating no substantial bias between the two methods. The narrow 95% limits of agreement (-0.122 – 0.102) further support the high level of agreement, suggesting these methods can be used interchangeably with minimal concern for systematic errors. This consistency might result from the fact that MLU_{CHS} and MLU_{Raw} rely on similar principles regarding the linguistic features in this task. For MLU_{PDP} and MLU_{CHS} , Table 5 shows a significant mean difference of 0.379, with MLU_{PDP} producing higher values than MLU_{CHS} does. This suggests that they capture different aspects of the linguistic data in utterance complexity or lexical diversity.

For the MCI group and healthy group, the measure MLU_{PDP} demonstrates fundamental deviation from other approaches. In the descriptive task, MLU_{PDP} again showed a significant negative bias against MLU_{Raw} , but no significant intergroup difference was observed ($p > .05$), and the 95% limits of agreement were relatively narrow. This pattern indicates that the descriptive task elicited less variability in pro-drop-related adjustments compared to the narrative task, and that the group difference observed in the narrative task was substantially decreased. One possible interpretation is that tasks imposing lower syntactic demands are less likely to reveal individual differences in grammatical processing; however, the specific cognitive mechanisms underlying this task-dependent effect remain to be directly investigated.

4. Discussion

This study represents a significant methodological experiment into the application of MLU as a linguistic measure in older adults, specifically within the Chinese language context where standardized guidelines for utterance segmentation remain uncertain. Our study systematically addressed two fundamental methodological questions:

Table 3

Pearson correlation coefficients of the Picnic Scene.

	MLU_{Raw}	MLU_{PDP}	MLU_{CHS}
MLU_{Raw}	1.000		
MLU_{PDP}	0.948	1.000	
MLU_{CHS}	0.999	0.944	1.000

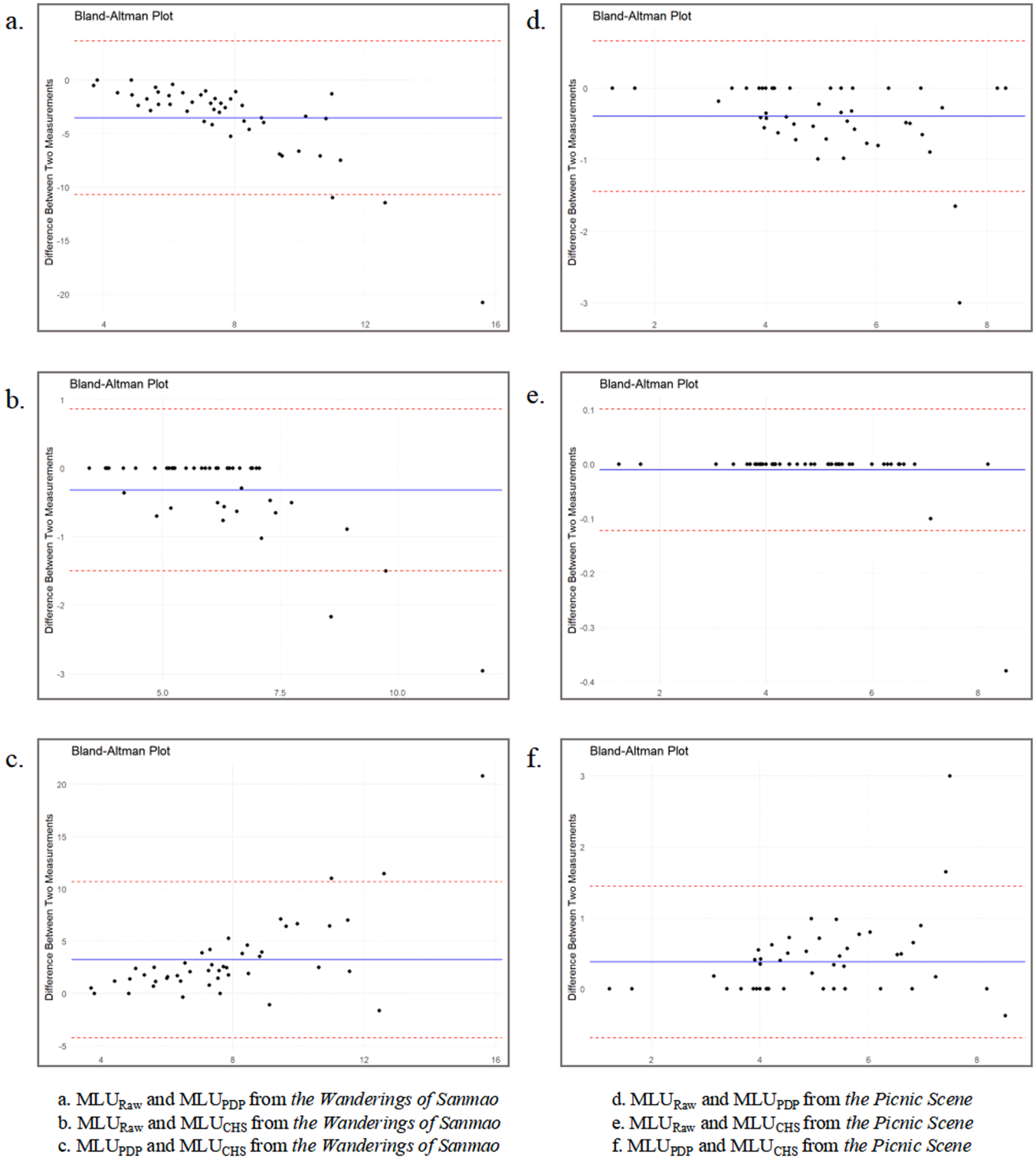


Fig. 3. The Bland-Altman difference plot.

how does the absence of strict guidelines affect the consistency of utterance segmentation judgments and how do different clause segmentation schemes impact the performance of MLU, specifically those addressing PDP and CHS which may influence the performance and reliability of MLU as a measurement tool. The findings reveal complex interactions between task demands, linguistic phenomena, cognitive condition, and methodological choices, demonstrating that different segmentation and editing procedures yield systematically different MLU

values. These results underscore the sensitivity of MLU to preprocessing decisions and highlight the importance of standardizing calculation procedures, rather than establishing that any single variant is more or less reliable as a measurement tool.

This study statistically analyzed the segmentation accuracy results of the *Wanderings of Sanmao* and the *Picnic Scene* tasks to answer the first research question. The findings revealed a slight difference in the overall consistency rates between the two tasks, with the narrative task at

Table 5
The Bland-Altman Analysis of the *Picnic Scene*.

Bland-Altman Analysis (the <i>Picnic Scene</i>)			
	MLU _{Raw} & MLU _{PDP}	MLU _{Raw} & MLU _{CHS}	MLU _{PDP} & MLU _{CHS}
Samples	47		
Mean (1 st method)	4.852	4.852	5.241
Mean (2 nd method)	5.241	4.863	4.863
Mean (Difference)	−0.389	−0.01	0.379
S. D. (Difference)	0.537	0.057	0.546
95% CI	−0.546 ~	−0.027 ~ 0.007	0.218 ~ 0.539
(Mean of Difference)	−0.231		
95% CI (Difference)	−1.441 ~ 0.663	−0.122 ~ 0.102	−0.692 ~ 1.449
<i>t</i>	−4.965	−1.228	4.751
<i>p</i>	< .001	0.226	< .001
Coefficient of Repeatability	1.29	0.112	1.293

42.17% and the descriptive task at 44.52%. These low accuracy rates between annotators provide empirical evidence regarding the impact of guideline ambiguity on utterance segmentation consistency. This aligns with Allen and Dench (2015), who noted that MLU calculation must be based on clear definitions of utterances; otherwise, it will lead to inconsistencies in calculations across different studies. These values, considerably lower than conventional reliability thresholds in linguistic annotation, underscore the inherent subjectivity in determining utterance boundaries when explicit operational criteria are absent. The substantial individual variability ranging from 14.29% to 71.43% in the narrative task further emphasizes how segmentation decisions become increasingly subject to individual interpretation without standardized rules. This variability manifested not only between participants but also within individuals across tasks, as illustrated by a participant who demonstrated a 15.56% accuracy differential between narrative and descriptive contexts. Such fluctuations suggest that task characteristics interact with annotators' internal linguistic frameworks to produce divergent boundary judgments. The finding that the descriptive task elicited marginally higher agreement than the narrative task aligns with theoretical expectations about scene description tasks activating a more stable linguistic scheme due to their concrete visual information and reduced demands for temporal integration.

Moreover, the narrative structure of the *Wanderings of Sanmao* requires more sophisticated connectivity and logical tracking than the *Picnic Scene*, which alone may increase ambiguity in boundary placement. This interpretation finds support in the observation that participants with suspected working memory limitations exhibited greater cross-task inconsistency, implying that higher cognitive load and low cognitive conditions can exacerbate segmentation inconsistency in the absence of clear guidelines, which is consistent with Marquardt's (2016) and Freeman's (2015) studies that cognitive decline can influence syntactic complexity. Despite overall performance, significant individual differences were observed, indicating that task type or content may influence ACC. The analysis posits that the *Picnic Scene* task, being a concrete scene description task, might activate a more stable language template, whereas the *Wanderings of Sanmao* task demands higher semantic integration, leading to performance variations in some participants. One plausible direction for future research is to examine whether higher cognitive load exacerbates segmentation inconsistency in the absence of clear guidelines, as suggested by previous studies linking cognitive decline to syntactic complexity (Marquardt, 2016; Freeman, 2015; Huang & Che, 2023; Huang, 2024; Yih et al., 2025). Thus, segmentation accuracy is influenced by both task and individual factors, which is in consistency with Klee and Fitzgerald's (1985) conclusion in children's research, and multiple designs may be employed to clarify its cognitive mechanisms.

Moreover, the differences observed between the two datasets suggest that the applicability of various MLU calculation methods may vary

depending on the language being studied or the specific research questions being addressed. Specifically, in the *Picnic Scene* dataset, the PDP method may introduce variability in MLU measures. This suggests that researchers should carefully select and validate their MLU calculation methods based on the specific characteristics of their data and the goals of their study. Different methods of MLU calculation may yield varying results, especially when working with data that has complex syntactic structures or when using probabilistic parsing techniques. Thus, studies should document the calculation method used to allow for accurate comparisons across different research findings.

The differential performance of the three MLU calculation methods (Raw, PDP, CHS) across tasks and groups reveals fundamental insights about linguistic phenomena that compromise measure validity. The Bland-Altman analysis demonstrated systematic methodological divergences that varied according to task complexity. In the cognitively demanding narrative task, MLU_{PDP} measure introduced substantial measurement bias with the mean difference between MLU_{Raw} and MLU_{PDP} reaching −3.529 (95% CI: −4.620 to −2.438; *p* < .001). The wide 95% limits of agreement (−10.731–3.673) indicate that for any given individual, MLU_{PDP} could be as much as 10.73 units lower or 3.67 units higher than MLU_{Raw}. This variability was significantly moderated by cognitive status, reflecting grammatical integration impairments associated with working memory deficits. This cognitive effect can be used as a potential biomarker as its clinical utility still remains upon standardized segmentation protocols.

Conversely, the relationship between MLU_{Raw} and MLU_{CHS} methods demonstrated remarkable stability across both tasks and cognitive groups. In the narrative task, their Bland-Altman analysis revealed a minimal mean difference of −0.316 with narrow limits of agreement and excellent repeatability, while in the descriptive task, agreement approached perfection with a negligible mean difference of −0.01 and exceptionally tight limits. This finding, reinforced by the Pearson correlations (*r* = 0.968–0.999 across tasks), indicates that cohesive repetition phenomena exert limited influence on utterance length measurement in older adult speech. This finding has important practical implications that manually adjusting MLU_{CHS} scores is time-consuming and rarely worth the extra effort for basic language analysis, especially since skipping these edits does not significantly distort simple descriptions of speech. Both raw and adjusted MLU methods produce consistent, reliable results for fundamental language assessments in older adults. However, because neither method is sufficiently sensitive to detect subtle early language changes linked to cognitive disorders on its own, they are less useful for detailed diagnosis. This is consistent with Szatloczki's (2015) conclusion that a single language indicator is difficult to fully capture cognitive decline, suggesting that MLU needs to focus on combining indicators such as speech speed, pauses, and semantic fluency to more comprehensively detect cognitive-related language changes. At the same time, the study found that narrative tasks are more likely to expose cognition-related grammatical integration deficits due to their requirement for activating higher-level working memory and semantic integration. This aligns with Szatloczki's view that spontaneous speech or narrative tasks can better reflect early language impairment in AD, and provides supplementary evidence for the syntactic measurement methodology to be applied locally in detecting cognitive impairment among Chinese older adults.

The interdependence of tasks observed across all measures emphasizes the critical importance of task selection in assessments of language and ageing. Narrative and descriptive tasks activated distinctly different linguistic subsystems with corresponding implications for MLU validity. The *Wanderings of Sanmao* task amplified MLU_{PDP} measure while simultaneously revealing cognitively modulated divergence patterns absent in the *Picnic Scene*. Specifically, MLU_{PDP} adjustment in narratives produced significantly larger MLU values due to frequent clause merging at pro-drop sites, which is a phenomenon substantially more pronounced in the MCI group where working memory constraints likely impaired referential tracking. This grammatical integration deficit

shows MLU_{PDP}'s negative bias against MLU_{Raw} yet positive discrepancy towards MLU_{CHS}. In contrast, *the Picnic Scene* description suppressed MLU_{PDP}'s volatility, reducing the MLU_{Raw} and MLU_{PDP} mean difference to -0.389 , while simultaneously eliminating cognitive group differences in variability ($p > .05$). This pattern inversely verifies task complexity as the core moderator of MLU's sensitivity to grammatical phenomena that only high demand narrative contexts activate the working memory resources necessary for pronoun resolution, making MLU_{PDP} a task-specific indicator rather than a universal measure. Meanwhile, the minimal impact of MLU_{CHS} measure across both tasks suggests that cohesive repetition represents a surface-level phenomenon that can be reasonably disregarded in MLU calculation without compromising measurement integrity. This can be seen as a pragmatic consideration given the resource constraints typical of clinical settings. As Dai (2022) found in their research, the oral expressive ability of older adults in southern Taiwan did not decline significantly with age over a five-year period. However, educational level significantly influenced MLU performance in storytelling tasks, but not in personal narrative tasks. This further corroborates the findings of our study, which suggest that the consistency and sensitivity of MLU vary across different types of tasks.

The correlation analyses further illustrate the operational characteristics of each MLU measure. The consistently strong MLU_{Raw} and MLU_{CHS} correlations confirm their fundamental measurement congruence across diverse linguistic outputs. In contrast, MLU_{PDP}'s weak and unstable associations with other methods, especially with MLU_{CHS} in *the Wanderings of Sanmao*, reflect its capture of linguistic dimensions related to syntactic integration rather than the length of utterances. Thus, MLU_{PDP} may be seen as a specialized measure for assessing grammatical cohesion rather than a direct replacement for conventional MLU. Crucially, the correlations in *the Picnic Scene* across all method pairs ($r = 0.944\text{--}0.999$) demonstrate that methodological choices become less consequential when linguistic output is constrained by simple visual scenes. This convergence effect highlights how task simplification artificially minimizes grammatical variation that would otherwise challenge utterance segmentation, which may tend to ignore the very first linguistic phenomena that might serve as early cognitive markers. However, narrative tasks like *the Wanderings of Sanmao*, despite introducing measurement complexity, offer superior validity for detecting subtle language changes associated with cognitive decline.

Our findings carry several critical implications for MLU standardization in research for syntactic complexity to answer the second research question. First, the low agreement between annotators (ACC) confirms that utterance segmentation cannot rely on implicit linguistic intuition when applied to older adults, particularly those with cognitive impairment. The framework based on ACC rate implemented here provides a quantifiable methodology for establishing segmentation reliability. Second, the task-dependent performance of MLU measures necessitates careful methodological alignment with assessment objectives. MLU_{Raw} and MLU_{CHS} demonstrate adequate reliability for basic language profiling, while MLU_{PDP} offers enhanced sensitivity to grammatical integration deficits but requires narrative contexts to activate its discriminant potential. Third, the cognitive modulation of MLU_{PDP} establishes its unique position at the intersection of linguistic and cognitive assessment, though its clinical implementation requires rigorous standardization to distinguish genuine impairment effects from measurement artifacts. Finally, the negligible impact of cohesive repetition across all analyses suggests that manual MLU_{CHS} correction may represent an inefficient allocation of analytical resources in linguistic research on language and ageing.

Overall, the MLU calculation under different control conditions is highly correlated. MLU_{Raw} and MLU_{CHS} can stably reflect basic linguistic complexity, while MLU_{PDP} captures a distinct dimension of syntactic cohesion. The preprocessing adjustments for pro-drop and cohesive repetition did not significantly influence the core characteristics of MLU, showing that it is less affected by specific control conditions. If MLU values are highly convergent under different preprocessing conditions, it

may indicate that the indicator has a relatively single representation dimension of language complexity, and additional analysis is required in combination with other indicators. The relationship between MLU and other factors like age and educational level needs to be further tested.

5. Conclusions

This study fundamentally challenges the presumption of MLU as a universally applicable measure in syntactic complexity calculation, revealing it instead as a complex construct whose validity is dynamically negotiated through interactions between linguistic phenomena, task demands, cognitive resources, and methodological choices. The substantial variability introduced by pro-drop phenomena, particularly in narrative contexts and among cognitively impaired individuals, demands explicit theoretical and operational attention rather than analytical adjustments and calculation with no systematic solutions. Moreover, the stability of MLU_{Raw} and MLU_{CHS} measures across diverse conditions confirms their utility as core measures for foundational language assessment. Methodological variability in MLU calculation indicates that it is not merely a single-dimensional measure but also a rich source of linguistic and cognitive information. By quantifying these relationships through advanced statistical approaches like Bland-Altman analysis and integrating them with cognitive assessment frameworks, we propose that MLU can be developed into a sensitive, standardized indicator of language change in aging, one that may reflect linguistic complexity while providing clinically actionable insights guided by standardized utterance segmentation rules. Future progress requires collaborative development of evidence-based segmentation guidelines, task protocols optimized for grammatical elicitation, and analytical frameworks that transform methodological challenges into theoretically meaningful dimensions of assessment.

When interpreting these findings, several limitations need to be considered. The sample size of 50 participants and restricted demographic sampling in Shanghai may limit generalization, particularly regarding dialectal influences on grammatical phenomena. In addition, the manual transcription and sentence segmentation of experimental materials may lead to potential omissions or errors, and consistency comparisons relied on only two researchers.

Moreover, future research should expand to include longitudinal designs tracking MLU trajectory in progressive cognitive disorders, incorporate neuroimaging methods to correlate grammatical processing with brain activity, and develop computational approaches to automate MLU_{PDP} identification while preserving linguistic sensitivity. Introducing more automated tools could improve transcription and segmentation efficiency, while involving more researchers in data calculation would enhance consistency and reliability. Longitudinal and cross-linguistic studies should investigate whether certain methods are more suitable for languages with specific syntactic or morphological properties, aiming to establish guidelines for selecting the most appropriate method based on language type, data complexity, and research objectives. This necessitates collaborative development of evidence-based segmentation guidelines, task protocols optimized for grammatical elicitation, and analytical frameworks transforming methodological challenges into theoretically meaningful dimensions of assessment, potentially achieved through large-scale collaborative studies and large language model processing examining methodological variability across diverse linguistic contexts to develop consensus guidelines on applying different calculation methods. Only through such rigorous methodological groundwork can MLU fulfill its potential as a valid, reliable measure across the lifespan.

CRedit authorship contribution statement

Yang Yiran: Writing – review & editing, Writing – original draft, Formal analysis. **Huang Lihe:** Writing – review & editing, Supervision, Resources, Conceptualization. **Yih Tsy:** Writing – review & editing,

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Writing – review & editing.

Declaration of Competing Interest

The authors declare that they have no known competing financial interests or personal relationships that could have appeared to influence the work reported in this paper.

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Data availability

Data will be made available on request.

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